Response to the Editor

Revised Manuscript: "Evaluating Tool-Using Language Agents: Judge Reliability, Propagation Cascades, and Runtime Mitigation in AgentProp-Bench"

Anonymous (for review)

April 17, 2026

1 Overview of Changes

We thank the editor for the constructive feedback. The manuscript has been substantially revised:

- Renamed from AgentHallu-Bench to **AgentProp-Bench** (see Section 9 below).
- **Repositioned** around evaluation reliability as the primary contribution, with propagation and mitigation as evidence.
- Added 18 new 2024–2026 references from IR/IS and agent-hallucination venues.
- Added standalone sections for Research Objectives (Section 3), Dataset (Section 4), and Discussion (Section 7).
- Expanded from ∼5,200 to ∼9,500 words.
- Corrected a methodological issue in stage-indicator definitions (disclosed transparently in Section 9).
- Collected 100 stratified human labels to validate the ensemble judge.
- Re-ran the interceptor under a concurrent no-interceptor control.

2 Response to Editor Point 1: Abstract

"Your abstract does not highlight the specifics of your research or findings but contains too much background information. Some details of your research would be nice, for example, numbers addressing the sample, data, percentage improvement, etc."

Addressed. The abstract has been rewritten to 195 words, leading with specific numbers rather than background. It now includes: sample sizes (2,000 tasks, 2,300 traces, 9 LLMs, 100 human labels), key results ($\kappa = 0.049$ for heuristic, $\kappa = 0.432$ for ensemble, $r_{2,3} \approx 0.62$ human-calibrated propagation, -23 pp interceptor effect), and methodology (concurrent n=600 control). Background material was reduced to one sentence.

> See the revised Abstract (page 1). Word count: ∼170 (down from 247).

3 Response to Editor Point 2: Research Objectives

"There needs to be an explicit research objective(s) stated, preferably as a separate section."

Addressed. A standalone Section 3 ("Research Objectives") now presents three explicit objectives:

RO1 Quantify agreement between heuristic, LLM-ensemble, and human judges on tool-using agent outputs.

RO2 Characterize parameter-level error propagation through agent pipelines under validated evaluation.

RO3 Evaluate a runtime interceptor under a concurrent control.

Each objective is accompanied by a rationale paragraph.

See Section 3 of the revised manuscript.

4 Response to Editor Point 3: Dataset Description

"I have questions about your dataset. So, you need to explicitly provide a description of your dataset(s), in the manuscript, which is currently lacking."

Addressed. Section 4 ("Dataset: AgentProp-Bench") now provides a full description including:

- Task design and construction methodology (Section 4.1)
- Domain breakdown with Table 1 showing 2,000 tasks across calendar (667), weather (674), medical (559), and knowledge (100) domains, plus 300 held-out retail tasks (Section 4.2)
- Tool schema and simulated environment (Section 4.3)
- Error-injection protocol with a worked example trace (Section 4.4)
- Trace collection details: 2,300 traces across 9 models (Section 4.5)
- Human label collection: 100 stratified labels, annotator description, sampling rationale (Section 4.6)
- Data availability with repository structure (Section 4.7)

See Section 4 and Table 1 of the revised manuscript.

5 Response to Editor Point 4: Discussion and Implications

"You must more clearly highlight your research's theoretical and practical implications; it is unclear how this research differs from existing work. There should be an explicit section addressing the discussion of results and implications."

Addressed. Section 7 ("Discussion and Implications") is now a dedicated section with four subsections:

- Section 7.1: Theoretical implications—rejection and recovery as orthogonal axes of agent robustness; why heuristic evaluation systematically misranks models.
- Section 7.2: Practical implications for agent designers—schema validators, pre-execution semantic checks, interceptor applicability.
- Section 7.3: Practical implications for IR/IS evaluation—substring heuristics are chance-level, LLM-ensemble judges need validation, methodology transfers to RAG/QA/agent benchmarks.
- Section 7.4: Explicit differentiation from AgentHallu (Liu et al., 2026), τ-bench, AgentBench, and BFCL.

See Section 7 of the revised manuscript.

6 Response to Editor Point 5: Copy Editing

"The manuscript requires thorough copy editing. Please keep in mind that many of the IP&M audience members are not native English speakers, so the manuscript must be accessible to them."

Addressed. The manuscript has been thoroughly copy-edited with the following guidelines applied:

- Sentences longer than 25 words were split.
- Idioms and metaphors were replaced with direct language.
- Every acronym (LLM, EPS, IR, etc.) is defined on first use.
- Active voice is used where possible.
- Each paragraph addresses one idea in 3–6 sentences.

7 Response to Editor Point 6: Manuscript Length

"The manuscript is short, so you have space to go into more detail concerning this research's prior literature, methods, data, and implications."

Addressed. The manuscript has been expanded from approximately 5,200 words to approximately 9,500 words. Major additions include:

- Expanded Related Work (Section 2): 6 thematic subsections with 18 new references.
- New standalone sections: Research Objectives (Section 3), Dataset (Section 4), Discussion (Section 7), Limitations (Section 8).
- New results: Rejection vs. Recovery decomposition (Section 6.3), human-validated judge reliability (Section 6.1), concurrent-control interceptor evaluation (Section 6.4).

8 Response to Editor Point 7: References and Literature Review

"Your references need to tie this research into the computing and information science community … you must update your literature review."

Addressed. The Related Work section (Section 2) has been reorganized into six thematic subsections and expanded with 18 new 2024–2026 citations. These include:

- IR/IS venues: at least 3 papers from SIGIR, CIKM, TOIS, and IP&M.
- Agent-hallucination work: AgentHallu (Liu et al., 2026) with an explicit differentiation paragraph (Section 2.3).
- LLM-as-judge methodology: recent work on judge reliability in IR evaluation contexts.
- Runtime monitoring and self-consistency approaches.

All references follow APA formatting via `elsarticle-num-names.bst`.

See Section 2 and the updated `refs.bib`.

9 Response to Editor Point 8: Author Guidelines and IP&M Style

"Please review recent IP&M articles on similar topics to get some insights on manuscript structure, presentation, and level of research."

Addressed. We reviewed recent IP&M articles on LLM evaluation, information retrieval, and agent benchmarking. The revised manuscript adopts:

- The `elsarticle` document class per IP&M author guidelines.
- A section structure consistent with empirical evaluation papers in IP&M (Introduction, Related Work, Objectives, Dataset, Methodology, Results, Discussion, Limitations, Conclusion).
- APA-style references.
- Explicit research objectives and a dedicated limitations section.

10 Additional Methodological Corrections Identified During Revision

During revision we identified that the original stage-indicator definitions in the submitted manuscript coupled S_2 (tool execution with injected parameters) and S_3 (final answer wrong), producing a definitionally forced $r_{2,3} = 1.00$. Specifically, the `classify_stage()` function in the analysis code mapped any row with `reached_output=True` to stage 3, and then computed S_2 as `stage` $\geq$ `2`, which included those same rows. This made $p_{S_2} = p_{S_3}$ for all models by construction.

We corrected the indicators to be genuinely independent: S_2 uses only the `eps` column (eps ≥ 2), and S_3 uses only the `final_correct` column. Under the corrected indicators, $p_{S_2} \neq p_{S_3}$ for most models (e.g., DeepSeek-V3: $p_{S_2} = 7.0\%$ vs. $p_{S_3} = 29.5\%$), and $r_{2,3}$ ranges from 0.43 to 0.97 across models (human-calibrated mean ≈ 0.62).

Additionally, we re-judged all 1,350 P2 semantic-wrong traces with the validated 3-LLM ensemble judge (rather than the substring heuristic used in the original submission), collected 100 stratified human labels to validate the ensemble at $\kappa = 0.432$ vs. humans, and re-ran the runtime interceptor under a concurrent n=600 no-interceptor control (rather than comparing against a historical baseline from a different task pool).

All numerical claims in the revised manuscript reflect the corrected analysis. See Section 5.2 for the revised methodology and the repository's `CORRECTED_CLAIMS_DIFF.md` for a one-to-one mapping between original and corrected numbers.

We also renamed the benchmark from AgentHallu-Bench to AgentProp-Bench to reflect its focus on propagation measurement and to avoid confusion with the concurrently published "AgentHallu" paper (Liu et al., 2026, arXiv:2601.06818), which does step-level attribution on a different dataset.

11 Summary of Changes

Metric	Original	Revised
Word count	∼5,200	∼9,500
References	28	46+
New 2024–2026 refs	0	18
Sections	5	9
Tables	6	8 + supplementary
Figures	1	5 + supplementary
Human labels	0	100